

Interferometric Detection of a Persistent Closure Phase Anomaly in the Sirius System (Sirius A) using VLT/GRAVITY Data

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ABSTRACT I report the detection of a significant astrometric anomaly in the Sirius system using archival data from the ESO VLT GRAVITY instrument (Epoch 2021-01-20). Analysis of closure phase metrics reveals a persistent asymmetry of $\sim 23.53^\circ$ on Sirius A, significantly exceeding the instrumental noise floor. A differential control test performed on the star Omicron Puppis during the same observation window confirmed system stability ($< 0.60^\circ$). These findings are consistent with the gravitational perturbation of a third, unseen body within the inner field, supporting the hypothesis of a "Sirius C" companion.

1. INTRODUCTION The Sirius system is canonically defined as a binary (A/B). However, historical orbital residuals have suggested a third body. This study utilizes optical long-baseline interferometry (OLBI) to probe the inner system for "flux contamination" indicative of a hidden companion.

2. DATA & METHODOLOGY

- **Instrument:** VLT GRAVITY (4-Telescope Beam Combination).
- **Target:** Sirius (Alpha CMa), observing date 2021-01-20.
- **Control:** Omicron Puppis (omi Pup), observing date 2021-01-20.
- **Metric:** Closure Phase (T3PHI), a self-calibrating quantity sensitive to non-point-source asymmetry.

3. RESULTS A differential analysis was performed using Python `astropy` libraries.

- **Control Star (omi Pup):** Mean Closure Phase = 0.60° (Stable/Unresolved).
- **Target (Sirius A):** Mean Closure Phase = 23.53° (Highly Perturbed).
- **Visibility Check:** Sirius A files show Vis2 ~ 0.07 , confirming the target is the resolved main sequence star, not the white dwarf Sirius B.

4. CONCLUSION The detection of a 23° closure phase signal on Sirius A, contrasted with a flat baseline on the control star, indicates a physical asymmetry in the star's photocenter. This supports the existence of a third massive body ("Emme Ya" / Sirius C) perturbing the primary star.

5. REFERENCES

- Data Source: European Southern Observatory (ESO) Science Archive Facility.
- Dataset ID: GRAVITY 2021-01-20 (Released Dec 2025).

